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An innovative machine learning based on feed-forward artificial neural network and equilibrium optimization for predicting solar irradiance

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As is known, having a reliable analysis of energy sources is an important task toward sustainable development. Solar energy is one of the most advantageous types of renewable energy. Compared to fossil fuels, it is cleaner, freely available, and can be directly exploited for electricity. Therefore, this study is concerned with suggesting novel hybrid models for improving the forecast of Solar Irradiance (I_s). First, a predictive model, namely Feed-Forward Artificial Neural Network (FFANN) forms the non-linear contribution between the I_s and dominant meteorological and temporal parameters (including humidity, temperature, pressure, cloud coverage, speed and direction of wind, month, day, and hour). Then, this framework is optimized using several metaheuristic algorithms to create hybrid models for predicting the I_s . According to the accuracy assessments, metaheuristic algorithms attained satisfying training for the FFANN by using 80% of the data. Moreover, applying the trained models to the remaining 20% proved their high proficiency in forecasting the I_s in unseen environmental circumstances. A comparison among the optimizers revealed that Equilibrium Optimization (EO) could achieve a higher accuracy than Wind-Driven Optimization (WDO), Optics Inspired Optimization (OIO), and Social Spider Algorithm (SOSA). In another phase of this study, Principal Component Analysis (PCA) is applied to identify the most contributive meteorological and temporal factors. The PCA results can be used to optimize the problem dimension, as well as to suggest effective real-world measures for improving solar energy production. Lastly, the EO-based solution is yielded in the form of an explicit formula for a more convenient estimation of the I_s .

Background

Recent advances in computational and engineering domains have provided reliable responses to various problems in human modern life^{1–5}. Intelligent tools, sophisticated simulation packages, and soft computing approaches are evident examples of these advances^{6–9}. In the field of energy, engineers have successfully employed these tools and methodologies to improve the sustainable development of renewable energy systems^{10–13}. Recently, solar energy has been introduced as an outstanding renewable source due to its numerous benefits like environmental friendliness, universality, high capacity, and inexhaustible supply^{14,15}. Scholars attempted to evaluate the solar energy production pattern by forecasting related parameters such as I_s . However, the evaluation of these

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